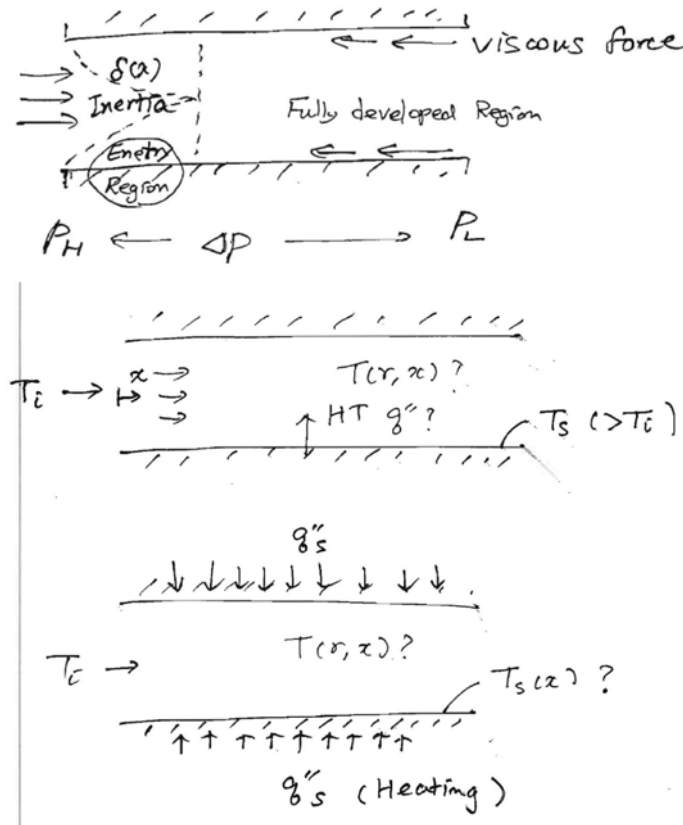


15. Introduction to internal flow (pipe flow)

Problem: Heat/momentum transfer by internal flow?



[Fluid mechanics]

External flow: Inertia ~ Viscous force

Internal flow: Pressure ~ Viscous force (Finite amount of inertia is soon used up and thus forcing (pumping) is needed to maintain flow)

→ Determining ΔP ("pressure drop") per unit length is important, i.e., $\frac{dp}{dx}$?

- Dimensionless pressure drop (Moody factor)

$$f \equiv \frac{-\left(\frac{dp}{dx}\right)D}{\frac{1}{2}\rho u_m^2} = \frac{(\Delta p)_D}{\frac{1}{2}\rho u_m^2} = \frac{\text{Pressure drop over a length } D}{\text{Dynamic pressure}}$$

u_m : mean velocity

(Note)

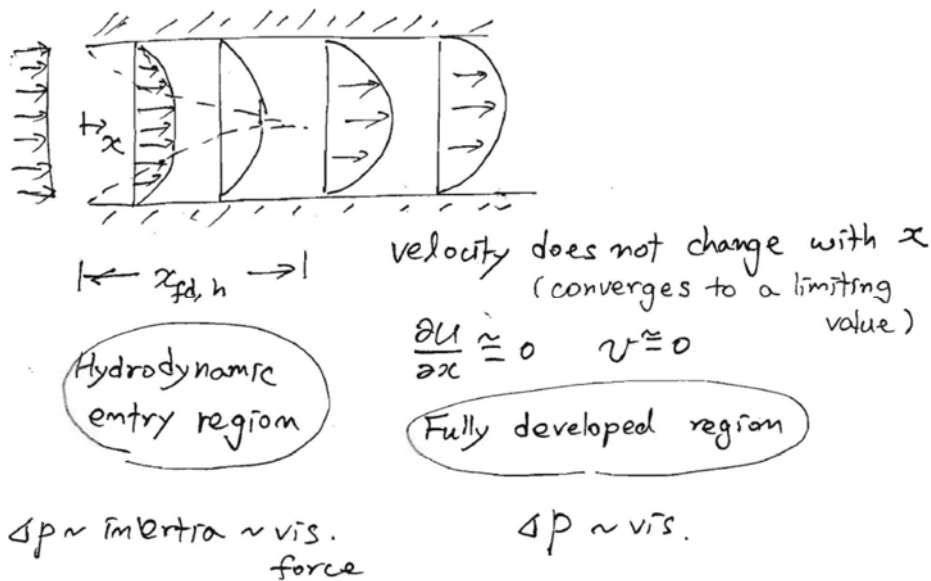
1. Pumping power $-w_s = \Delta p \frac{\dot{m}}{\rho} = \Delta p \dot{Q} \text{ [W]}$
 $\therefore -w_s = \dot{m}(h_e - h_i) = \dot{m}[(u + pv)_e - (u + pv)_i] \leftarrow 1^{\text{st}} \text{ law of thermodynamics}$
2. Also, $f = 4C_f$
3. Flow mode determined by $Re_D = \frac{u_m D}{\nu}$, NOT $Re_x = \frac{u_m x}{\nu}$ ($Re_{D,c} \cong 2300$)

Pipe diameter is more important than length (axial location x)

If the flow is fully developed, x is not important at all!

$$\frac{x_{fd,h}}{D} \cong 0.05 Re_D \text{ for laminar flow}$$

$$10 \leq \frac{x_{fd,h}}{D} \leq 60 \text{ (in this class } \frac{x_{fd,h}}{D} \cong 10) \text{ for turbulent flow}$$



4. Sketch variations of pressure and shear stress as a function of x

[Heat transfer]

- HT coefficient: $q'' = h\Delta T \rightarrow \Delta T? \rightarrow q'' = h(T_s - T_m)$ where T_m : mean temperature

<Def>

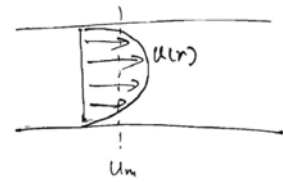
- Mean (average, bulk) velocity: $\dot{m} = \rho u_m A_c = \int_{A_c} \rho u dA \rightarrow u_m \equiv \frac{1}{\rho A_c} \int_{A_c} \rho u dA$

For a circular pipe, $u_m = \frac{2}{r_0^2} \int_0^{r_0} u r dr$

- Mean (average, bulk) temperature:

$$\text{Rate of energy transport } \dot{m} c_v T_m = \int_{A_c} (\rho u) c_v T dA$$

$$\text{For a circular pipe, } \dot{m} c_v T_m = \int_0^{r_0} (\rho u) c_v T \cdot 2\pi r dr \rightarrow T_m = \frac{2}{u_m r_0^2} \int_0^{r_0} u T r dr$$



(ex p.1)

$$\begin{cases} u = C_1 \\ T(r) = T_1 + C_2 [1 - (r/r_0)^2] \end{cases}$$

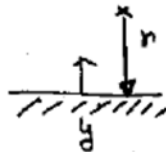


$$Nu_D = ?$$

$$Nu_D = \frac{h \cdot D}{k} = \frac{2h \cdot r_0}{k}$$

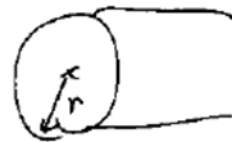
$$q'' = h(T_s - T_m)$$

$$q'' = -k \frac{dT}{dy} \Big|_{y=0}$$



$$= +k \frac{dT}{dr} \Big|_{r=r_0}$$

$$\frac{dT}{dr} = -\frac{2C_2 r}{r_0^2}$$



$$= -\frac{2kC_2}{r_0}$$

$$T_s = T(r_0) = T_1, \quad T_m = ?$$

$$T_m = \frac{2}{u_m r_0^2} \int_0^{r_0} u T r dr = \frac{2}{C_1 r_0^2} \int_0^{r_0} C_1 \{T_1 + C_2 [1 - (r/r_0)^2]\} r dr$$

$$= \frac{2}{r_0^2} \int_0^{r_0} \left\{ T_1 r + C_2 r - \frac{r^3}{r_0^2} \right\} dr = T_1 + \frac{C_2}{2}$$

$$h = \frac{q''}{T_s - T_m} = \frac{+k \frac{dT}{dr}}{T_s - T_m} = \frac{-\frac{2kC_2}{r_0}}{T_1 - T_1 - \frac{C_2}{2}} = \frac{4k}{r_0}$$

$$\therefore Nu_D = \frac{\frac{4k}{r_0} \cdot 2r_0}{k} = 8$$